

Chp 7.2: Addition and Subtraction Formulas**Addition and Subtraction Formulas**

Formulas for sine.

$$\sin(s + t) = \sin s \cos t + \cos s \sin t$$

$$\sin(s - t) = \sin s \cos t - \cos s \sin t$$

Formulas for cosine.

$$\cos(s + t) = \cos s \cos t - \sin s \sin t$$

$$\cos(s - t) = \cos s \cos t + \sin s \sin t$$

Formulas for tangent.

$$\tan(s + t) = \frac{\tan s + \tan t}{1 - \tan s \tan t}$$

$$\tan(s - t) = \frac{\tan s - \tan t}{1 + \tan s \tan t}$$

I. Evaluating Expressions Involving Inverse Trigonometric Functions

When evaluating expressions involving inverse trigonometric functions, remember that an expression like $\cos^{-1} x$ represents a(n) angle.

II. Expressions of the Form $A \sin x + B \cos x$

We can write expressions of the form $A \sin x + B \cos x$ in terms of a single trigonometric function using the Addition Formula for Sine. If A and B are real numbers, then

$$A \sin x + B \cos x = \underline{k \sin(x + \phi)}, \text{ where } k = \sqrt{A^2 + B^2}, \text{ and } \phi \text{ satisfies } \cos \phi = \frac{A}{\sqrt{A^2 + B^2}}$$

and $\sin \phi = \frac{B}{\sqrt{A^2 + B^2}}$

Example 1: Use an Addition or Subtraction Formula to find the exact value of the expression.

a. $\tan 165^\circ$

b. $\cos \frac{17\pi}{12}$

c. $\sin \left(\frac{-5\pi}{12} \right)$

Example 2: Use an Addition or Subtraction Formula to write the expression as a trigonometric function of one number, and then find its exact value.

a. $\cos 23^\circ \cos 67^\circ - \sin 23^\circ \sin 67^\circ$

b. $\sin \frac{3\pi}{4} \cos \frac{\pi}{4} - \cos \frac{3\pi}{4} \sin \frac{\pi}{4}$

c. $\frac{\tan 55^\circ - \tan 10^\circ}{1 + \tan 55^\circ \tan 10^\circ}$

d. $\cos \frac{5\pi}{18} \cos \frac{\pi}{9} + \sin \frac{5\pi}{18} \sin \frac{\pi}{9}$

Example 3: Prove the identity.

a. $\cos \left(x - \frac{\pi}{2} \right) = \sin x$

b. $\cos \left(x + \frac{\pi}{3} \right) + \sin \left(x - \frac{\pi}{6} \right) = 0$

c. $\tan \left(x + \frac{\pi}{3} \right) = \frac{\sqrt{3} + \tan x}{1 - \sqrt{3} \tan x}$

d. $\frac{\sin (x+y) - \sin (x-y)}{\cos (x+y) + \cos (x-y)} = \tan y$

e. $\cos(x + y)\cos(x - y) = \cos^2 x - \sin^2 y$

f. $\csc\left(\frac{\pi}{2} - u\right) = \sec u$

Example 4: Write the given expression in terms of x and y only.

a. $\cos(\sin^{-1} x - \tan^{-1} y)$

b. $\sin(\sin^{-1} x + \cos^{-1} y)$

Example 5: Find the exact value of the expression.

a. $\sin\left(\cos^{-1}\left(\frac{1}{2}\right) + \tan^{-1} 1\right)$

b. $\tan\left(\sin^{-1}\left(\frac{3}{4}\right) - \cos^{-1}\left(\frac{1}{3}\right)\right)$

Example 6: Evaluate each expression under the given conditions.

a. $\sin(\theta - \phi)$; $\sin \theta = \frac{4}{5}$, θ in Quadrant II, $\tan \phi = \frac{1}{3}$, ϕ in Quadrant III

b. $\cos(\theta + \phi)$; $\tan \theta = \frac{4}{3}$, θ in Quadrant III, $\cos \phi = \frac{2}{3}$, ϕ in Quadrant IV

Example 7: Write the expression in terms of sine only.

a. $-\sqrt{3}\sin x + \cos x$

b. $5(\sin 2x - \cos 2x)$